Radio Networks

The Model

Broadcast

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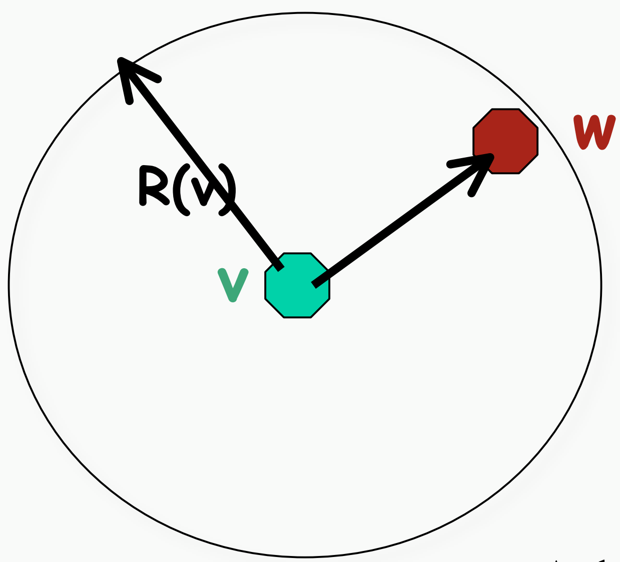
A **radio network** is a set of stations (nodes) located over a support Euclidean Space.

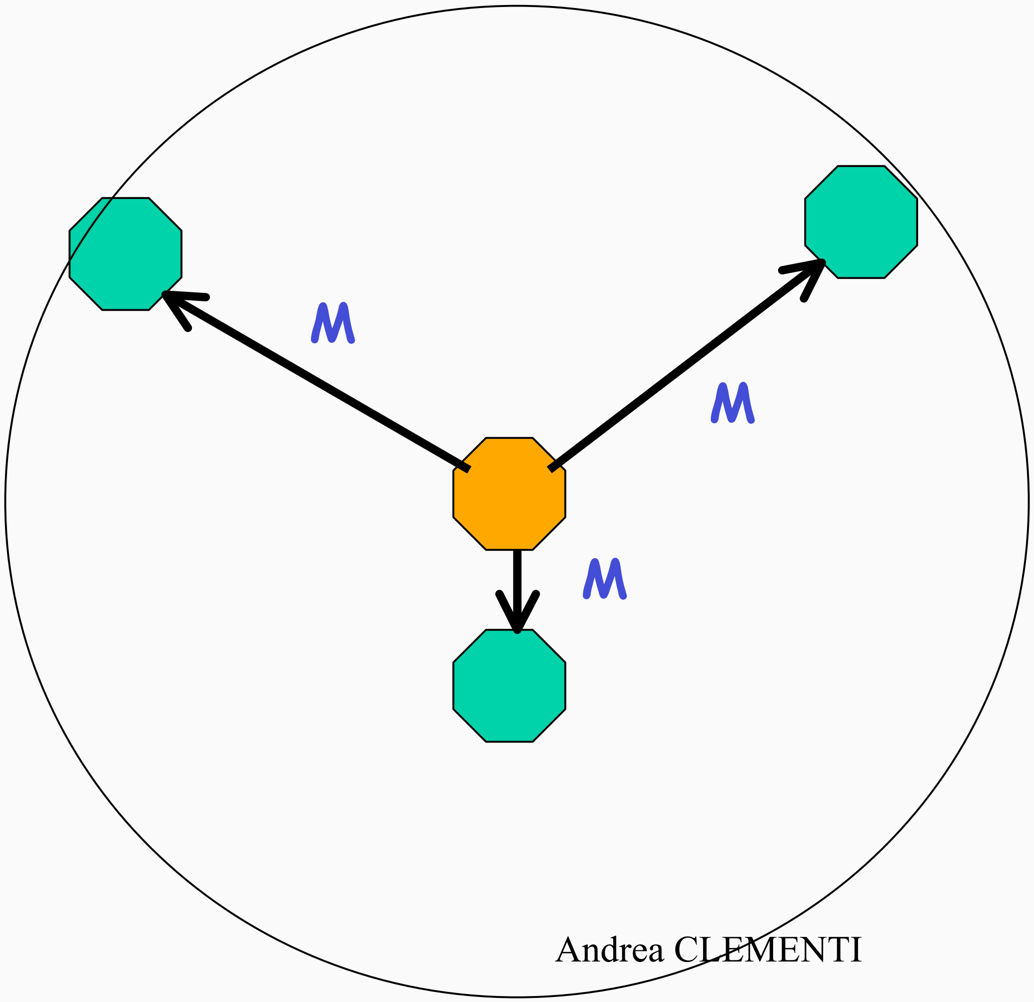
To each node **v**, a transmission range **R(v)>0** is assigned.

A node **w** can receive a *msg* M from **v** only if

$$d(v,w) <= R(v)$$



When a node **v** sends a msg M, M is sent over **all** the disk (Broadcast Transmission) in one TIME SLOT



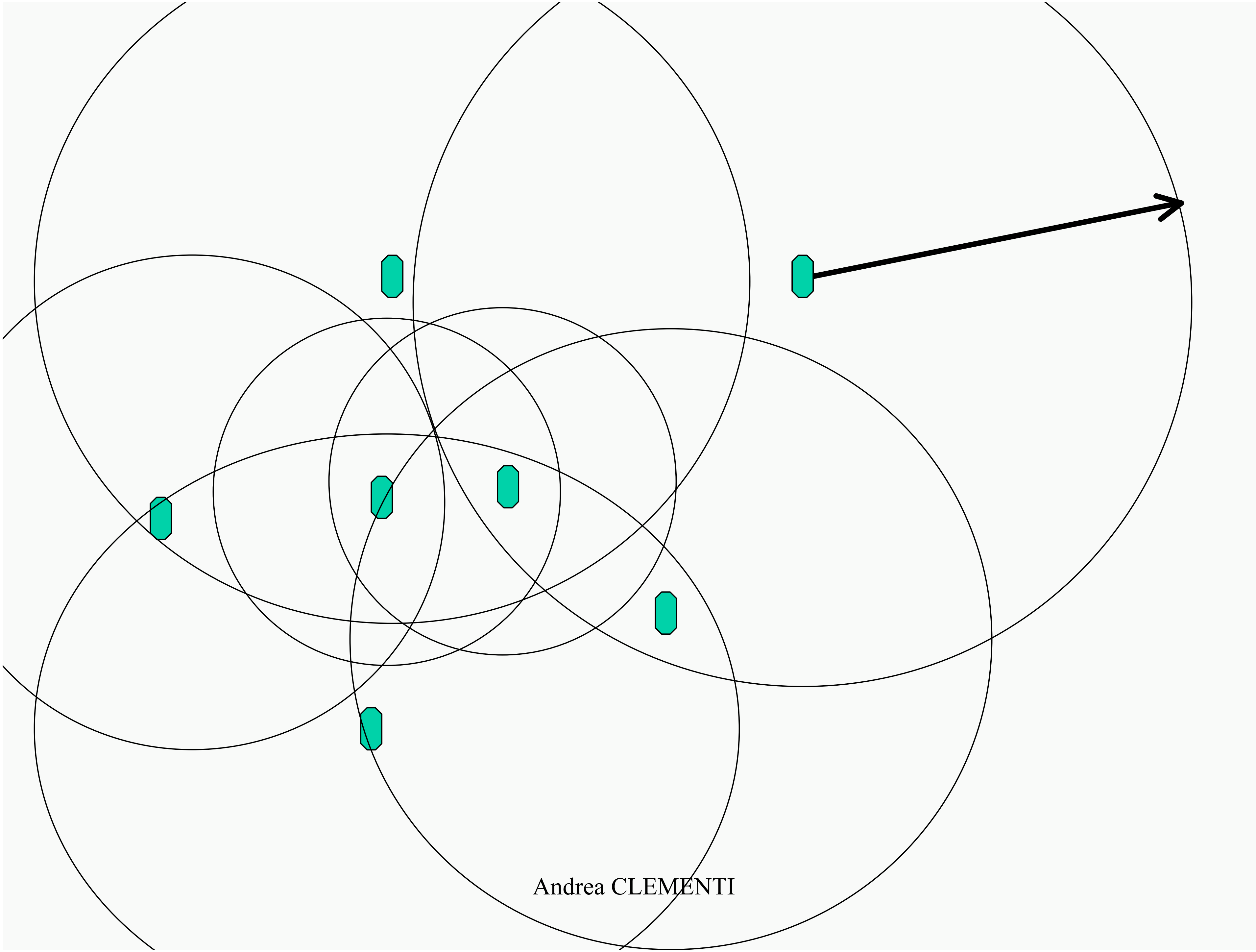
Radio Networks are SYNCHRONOUS SYSTEMS

All nodes share the same global clock. So,

Nodes act in **TIME SLOTS**

Message transmissions are completed

within **one time slot**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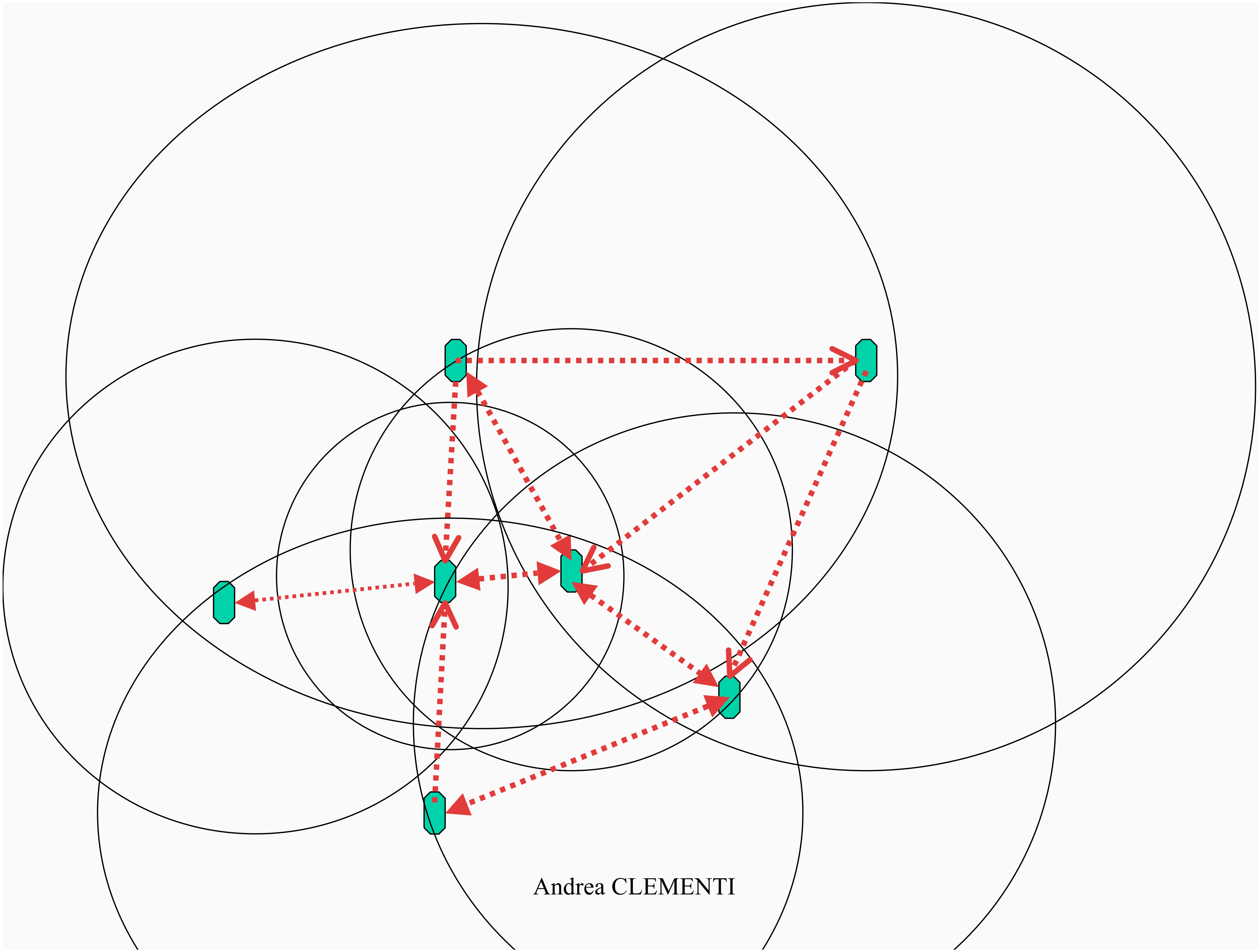

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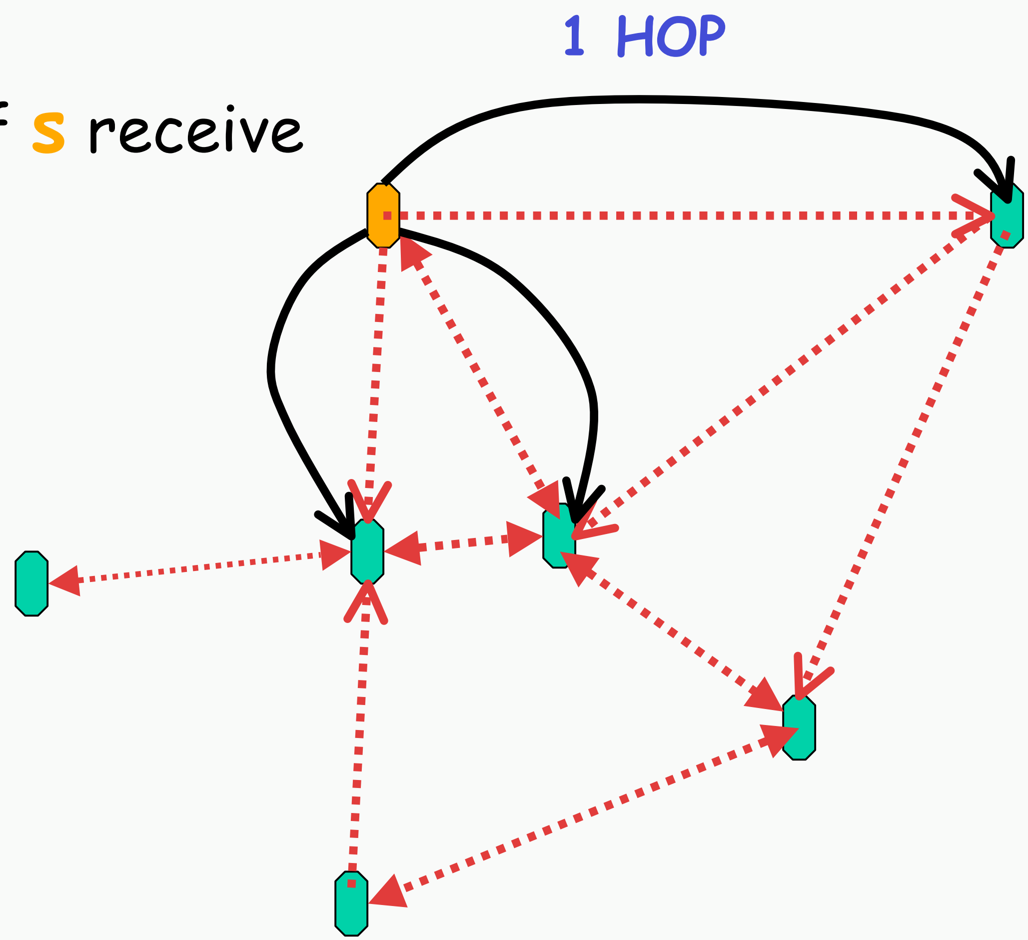
The Range Assignment uniquely determines a

Directed Communication Graph G(V,E)

All in-neighbors of s receive the msg in 1 HOP

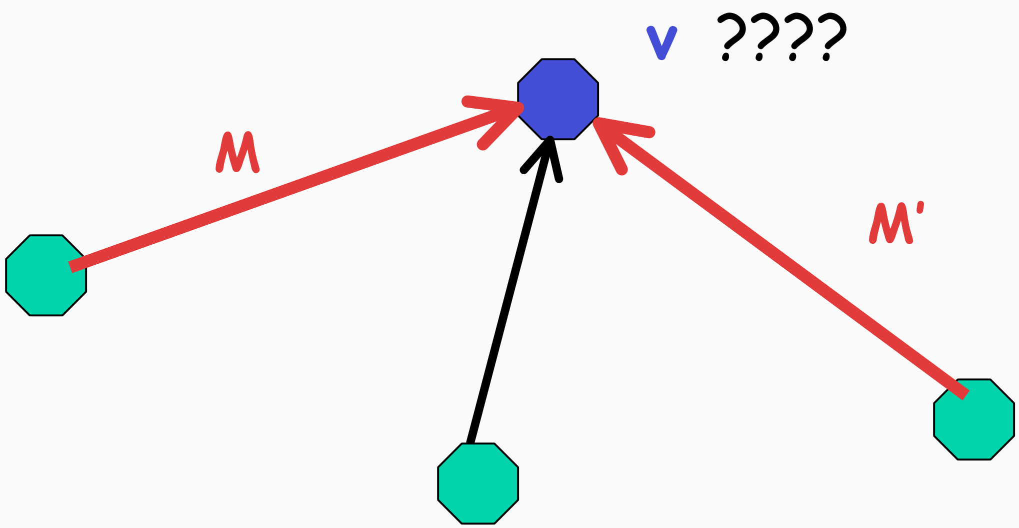
unless.....



MESSAGE COLLISIONS

If, during a time slot,
two or **more in-neighbors** send a msg to **v** THEN

v does not receive anything.



RADIO MODEL:

a node v receives a msg during time slot T

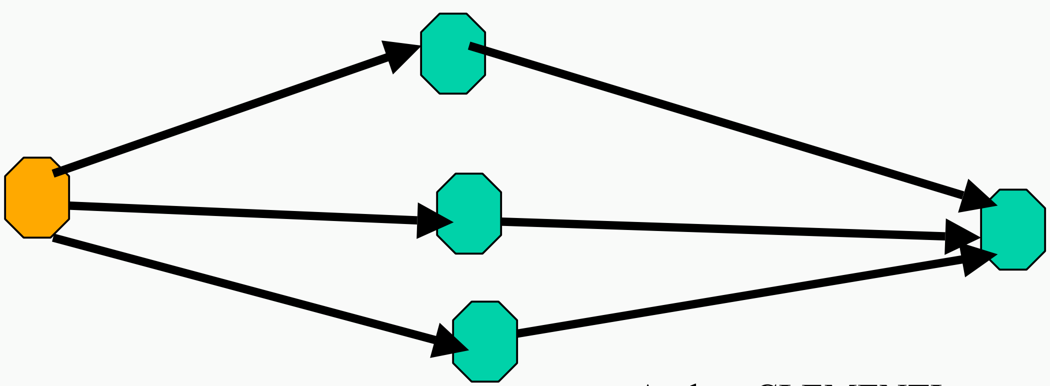
IFF

there is **exactly** **one** of its in-neighbors that sends a msg during time slot T

TASK:

BROADCAST OVER A RADIO NETWORK $G(V,E)$

NOTE: FLOODING DOES NOT WORK !!!!!



CORRECTNESS (Strongly-Conn. $G(V,E)$, source **s**) :

A Protocol **completes** Broadcast from **s** over ***G***
if **there is** one time slot s.t. every node is **INFORMED**
about the source msg.

TERMINATION

A Protocol terminates if there is a time slot **t** s.t.
every node **stops** any action WITHIN time slot **t**.

HOW can we **AVOID** MSG COLLISIONS ???

IDEA: **ROUND ROBIN !!!**

Start with Assumptions:

- nodes know a ***good apx*** of **$|V| = n$**

- nodes are indexed by **$0, 2, \ldots, n-1$**

then .....

IDEA 1: ROUND ROBIN !!!

Start with Assumptions:

- nodes know a ***good apx*** of $|V| = n$

- nodes are indexed by $0, 2, \ldots, n-1$

then .....

ROUND ROBIN PHASE

A ***Phase*** of ROUND ROBIN consists of **n** time-slots

At TIME T = 0,1,2,.....

- NODE **i**=T, if <u>informed</u>, sends the source msg;
- All the Others do NOTHING

What can we say **AFTER** one Phase of **RR** ?

Assume that label(s) = J (initially
J is the only informed one)

During the *FIRST PHASE* (*n time slots*):

Fact: ALL out-neighbors of s will be informed after the First PHASE.

No MSG Collision occurs...

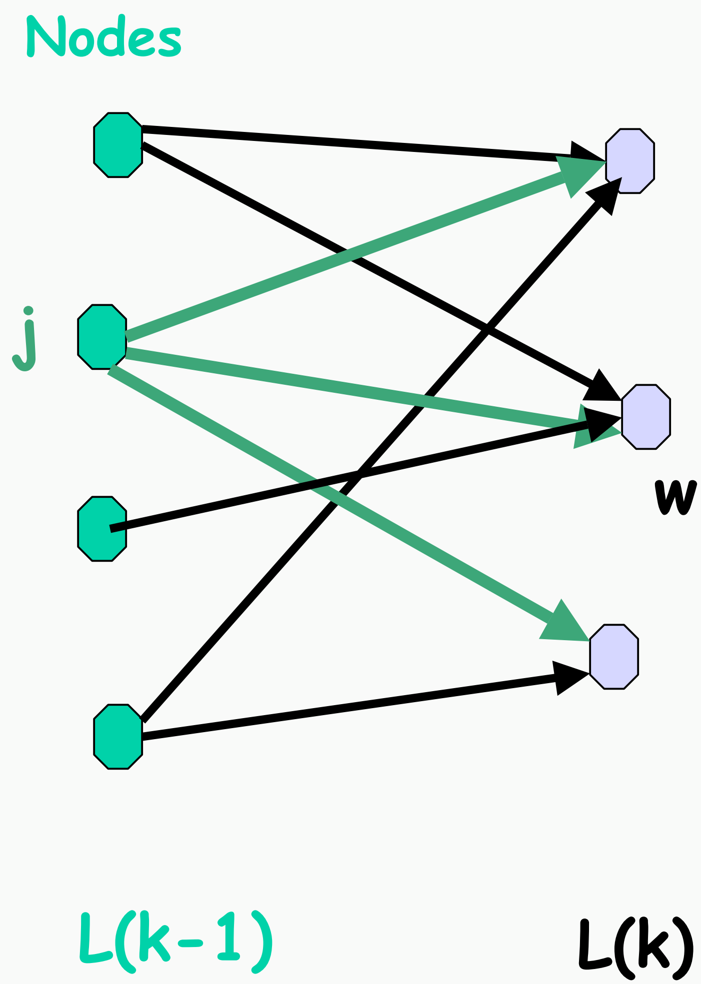
IDEA 2:

LET'S RUN THE **RR** PHASE FOR **L consecutive times**

THM. After Phase k, All nodes within Hop-Distance k from the source s

Proof. By induction on **HOP-DISTANCE = PHASE** k

Inductive Step: Phase k



at time slot j:

- j sends to all its out-neighbors **w**
- no others are active

So, ALL w's will receive the msg.

This Argument holds for all nodes in L(k-1).

So **all** nodes in **L(k)** will be informed after Phase **k**

Corollary (RR COMPLETION TIME).

Let D be the (unknown) **source eccentricity**. Then,

D **RR**-Phases suffice to INFORM all NODES

WHAT ABOUT **TERMINATION** ???

... It depends on the Knowledge of Nodes.

If they know **n** they CAN decide to stop... !

WHEN ????

The (unknown) source eccentricity is at most $n-1$,

so....

They all have the global clock ==> they all can decide to stop AFTER the RR Phase $n-1$

THM. Protocol RR

- completes Broadcast in $D \times n$
- terminates Broadcast in $O(n^2)$

Terrible question.....

What can we say if

<u>**NODES DO NOT KNOW**</u> any good bound on n

????

COMPLEX RESULTS:

-In **UNKNOWN** RADIO NETWORKS,

RR Completes in $O(D\ n) = O(n^2)$ time slots
Termination ?????

-There is an **optimal** Protocol that **completes** in

$$O(\ n \log^2 n\ )\ \text{time slots}$$

OBS.

RR does not exploit **parallelism** *at all*

GOAL:

SELECT PARALLEL TRANSMISSIONS

A "selective" method.

DEF. Given $[n] = \{1,2,...,n\}$ and $k <= n$,
a family of subsets

$$\mathcal{H} = \{H_1, H_2, ..., H_t\}$$

is **(n,k)-selective** if for *any* subset $S < [n]$ s.t.
$|S| <= k$, an $H < \mathcal{H}$ exists s.t.

$$|S \cap H| = 1$$

Trivial Fact.

The family $H = \{\{1\},\{2\},\ldots,\{n\}\}$ **is (n,k)-selective for any k.**

How a selective family can be used to BROADCAST ?

Restriction: Nodes know **n** and **d**;

(**As for the ***completion*** time: they can be removed)

SET UP:

All nodes know the same **(n,d)-selective** family

$$\mathcal{H} = \{H_1, H_2, \ldots H_i, \ldots . H_t\}$$

where

$$d = \text{max-degree}(G)$$

Protocol SELECT1.

- Protocol works in consecutive Phases J=1,2,... (as ***RR*** !!!).

-At time slot **i** of every Phase,

every informed node in H_i transmits

Protocol Analysis.

- **Lemma 1**. After Phase j, all nodes at distance at most j will be informed.

Proof. By induction on j. j=1 is trivial. Then, consider a node y at distance j. Consider the node subset

N(y)={z < V| z is a neighbor of x & z is at distance j-1}

Since N(y) < [n] and |N(y)| <= d, apply **(n,d)-selectivity** and get **the thesis**.

Is it correct?

NO!!!!

We are not considering the impact of **informed** nodes z in level j during phase j !

a) if you put z into $N(y)$, z could be selected but not already informed

b) if you don't put z into $N(y)$, z could be informed and create **collisions**

So what?

A very simple change makes the protocol correct!!!

ONLY NODES THAT HAVE BEEN INFORMED DURING PHASE j-1
WILL BE ACTIVE DURING PHASE J

No unpredicatble collisions and enough to inform level j

Lemma 1 is now true!, so after D phases, all levels will be informed.

Completion time is $O(D\ |\mathcal{H}|)$

So we need minimal-size selective families.

THM (ClementiMontiSilvestri 01).
For sufficiently large **n** and **k<=n**, there exists an (n,k)-selective family of size

O(k log n) and this is optimal !

If we plug-in the minimal size **(n,d)-selective** family into the protocol, we get:

$O(D\ d \log n)$ time

So if **D** and **d** are both small (most of "good" networks), we have a much better time than the **RR** one

THE LOWER BOUND.

Can the selective protocol be improved for general graphs?

NO!

THM. In **directed general graphs**, the use of a **selective family** is somewhat **necessary**,

GET for $Dd < n$: $\Omega(D\ d\ \log(n/D))$

LOWER BOUND.
Construct a Layered Directed Network.
$L0 = \{s\}$, then Lj as follows:

Let m < min size (n/D,d)-selective family.
Adv chooses the next level by looking at **Prot**'s transmissions for the next m **time slots**
as if Lj was ALL the rest of nodes.

He then chooses the subset of nodes **not selected** by **Prot** **(since m < min size (n/D,d)-selective).**

This subset becomes Lj

OBS.

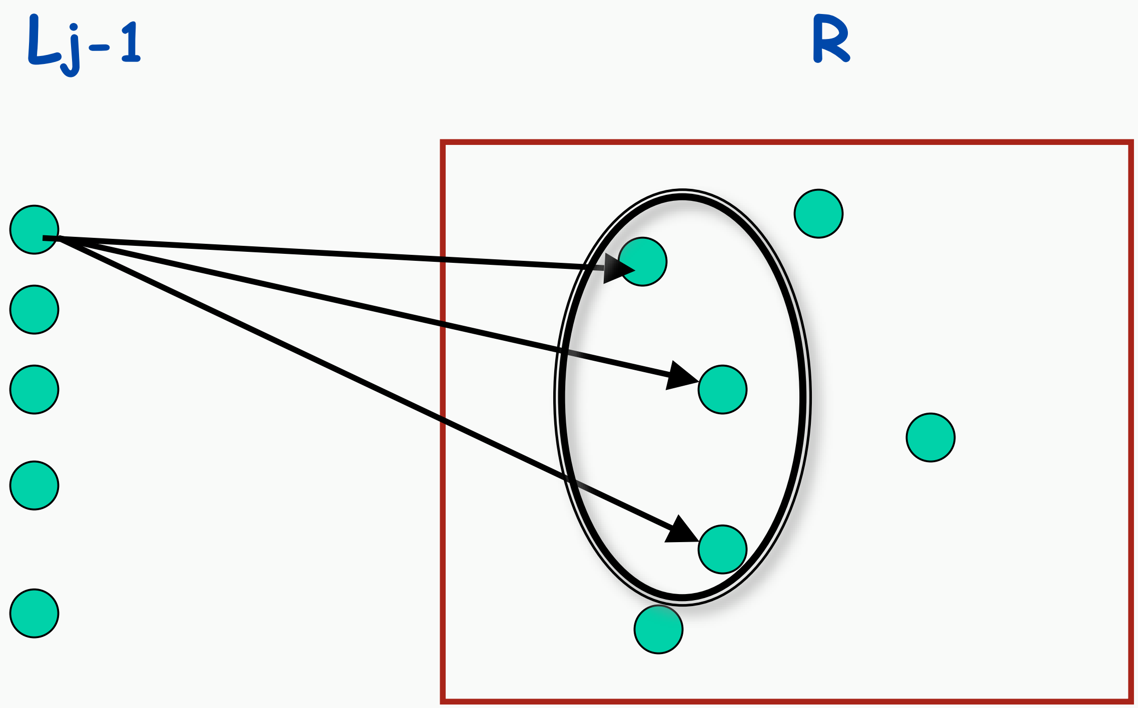
- ***Adv*** can do this for **$O(n/D)$** levels in order to produce a network of diameter **D** still keeping $|R| > n/2$.

- The behaviour of **Prot** is the **same** in both scenarios:

R = ALL THE REST OF NODES

R = LJ

THE LOWER BOUND (Proof).



Bipartite Complete Graph between Lj-1 and the **unselected** subset of R

Proof (LOWER BOUND).

-The Layered Graph shows that, in order to inform each Level, **Prot** needs to produce a
transmission scheduling

$$H = \{H_1, \ldots, H_k\}$$

which **must** be

(n/D, d)-selective.

So $|H|$ must be $\Omega(d \log(n/D))$ **and globally get**

$\Omega(D * d \log(n/D))$ **time.**

Random vs Deterministic: an Exponential Gap

Lower Bound for deterministic protocol when

$d = n$ and **$D = 3$** --> **$\Omega(n \log n)$**

What about **Randomized** Protocols ?

Example: at every time slot,

every **informed** node transmits with probability **1/2**.

The **BGI** RND Protocol
(Case of **d-regular** layered graphs (as in the L.B))

Repeat for K = 1,2,.... (Stage)

Repeat for j = 1,2, ..., c log n

If node x has been *informed* in Stage k-1
then x transmits with probability

$1/d$

Protocol Analysis.

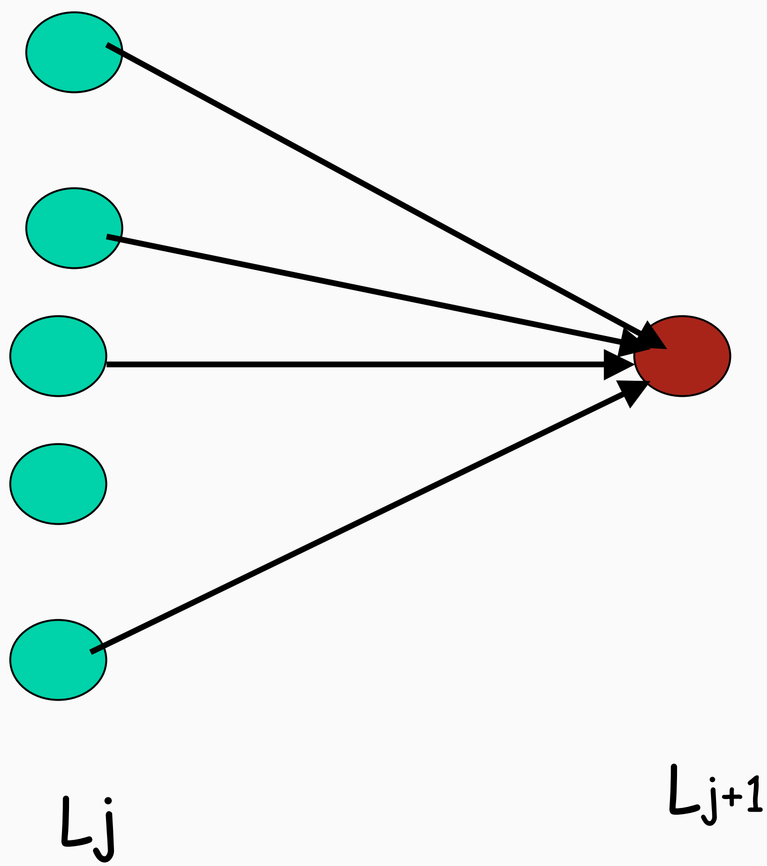
THM. Prot. BGI completes Broadcast within
O(D) Stages, so within

O(D log n) time step

WITH HIGH PROBABILITY

PROOF. By Induction on Level L=1….D.

D=1 --> Trivial. So assume all nodes of Lj are informed after **t = O(j log n)** time slots. Consider STAGE **j+1 .**



Which is the **Prob** that y will be informed during STAGE **J+1**?

Probability in **1 time slot**:

$d * (1/d) (1-1/d)^{d-1} > 1/8$

Probability that he is **not** informed
in (1 Stage =) $c \log n$ **independent** time slots:

$< (1-1/8)^{c \log n} < e^{- c/8 \log n} < 1/n^{c/8}$

since

- **Independent *rnd*** choices
- $(1-x) < e^{-x}$ for any $0<x<1$

we need this for ***all*** nodes $(< n)$

apply ***UNION BOUND*** ***twice:***

* *Pr(∃ BAD node)* $< n\,(1/n^{c/8}) < 1/n^{c-1/8}$

we need this for $k = D < n$ Stages

** *Pr(∃ BAD Stage)* $< 1/n^{c-2/8}$

By choosing $c > 10$, you get Theorem

WITH HIGH PROBABILITY = (1-1/n)

(*) Task:

Extend the BGI Protocol to

General Graphs

So to complete Broadcast in

$O(D \log^2 n)$ time slot (W.H.P.)

Restriction: nodes know **n**

You are interesting in learing more?

See the paper (CMS01.pdf)
in the Course Web Page

Thanks!

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